

**SPEECH-BASED REFLECTIVE SUPPORT MODULE
FOR STUDENT STRESS MONITORING**

25-26J-480

Project Proposal Report

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Engineering

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DECLARATION

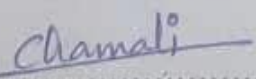
We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

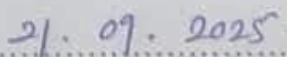
Name	Signature	Date
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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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Date

ABSTRACT

Academic stress poses significant challenges to university students, affecting mental health, learning outcomes, and overall well-being. This report presents a Self-Monitoring and Anonymous Counselling Digital Companion designed to help students manage stress through voice journaling and AI-driven emotional analysis. The application allows students to record voice entries, detect stress-related keywords, and classify emotional states using machine learning and deep learning models, including KWS and CNN. Synthetic voice data is incorporated to augment training and improve model accuracy. The system generates personalized emotional mind maps and delivers AI-guided coping strategies, relaxation exercises, and access to a moderated peer-support community. Developed with React Native for cross-platform mobile use and Firebase for secure backend services, the application ensures real-time data management and privacy. Evaluation focuses on model performance, user engagement, and effectiveness in stress reduction. This integrated approach enhances students' self-awareness, emotional regulation, and resilience, offering a practical and interactive solution for sustainable academic stress management.

Keywords- Keyword Spotting (KWS), Convolutional Neural Networks (CNN), Artificial Intelligence (AI), Mel-Frequency Cepstral Coefficients (MFCC)

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LIST OF ABBREVIATIONS

Abbreviation	Description
CNN	Convolutional Neural Networks
AI	Artificial Intelligence
MFCC	Mel-Frequency Cepstral Coefficients
DL	Deep Learning
ML	Machine Learning
TTS	Text-to-Speech
KWS	Keyword Spotting
TESS	Toronto Emotional Speech Set
SVR	Sentimental Voice Recognition

1. INTRODUCTION

Stress has emerged as a defining concern for contemporary youth, with university and college students exhibiting some of the highest reported levels. The need to manage simultaneous workloads, interpersonal commitments, and individual adversities keeps the physiological and psychological stress systems activated. Formulating and internalizing sustainable stress-regulatory strategies is, therefore, a university-level imperative: such strategies protect not only the pursuit of Grade Point Average objectives but also the body and mind from cumulative damage, given that heightened alertness is now the backdrop of nearly every educational day [1]. Existing studies consistently show a strong connection between academic burnout and how well students can manage stress. When exposed to academic pressures, students often experience a range of psychological and physical responses, such as increased anxiety, constant fatigue, loss of motivation, frequent headaches, digestive problems, and ongoing sleep difficulties. Research also highlights that resilience plays an important role in helping students regulate emotions and cope with the demands of stressful learning environments. Since individuals respond to stress differently, adopting effective coping strategies is essential for healthy adjustment [2].

To mitigate the growing incidence of academic distress among students, we propose the development of a mobile application specifically designed for stress management, delivering psychological support in a manner that is both interactive and intuitively navigable.

Key Components:

- i. **User Management:** Profile creation, mood tracking, and weekly emotional summaries.
- ii. **Emotion Tracking:** Daily check-ins, coping suggestions, and AI-generated soundscapes for relaxation.
- iii. **Stress Heatmap:** Predicts high-stress periods by analyzing mood, workload, sleep, and screen time.
- iv. **Community Forum:** Anonymous, moderated space for peer support.
- v. **Voice Journaling:** Detects emotional tones from speech and visualizes patterns.
- vi. **AI Chatbot:** Provides supportive conversations, coping strategies, and counselor alerts if needed.
- vii. **Relief Tools:** Includes breathing exercises, motivational content, mini-games, and readings.

- viii. **Task & Reward System:** Helps manage tasks and encourages consistency through rewards.
- ix. **Emergency Support:** Direct access to counselors or guardians during distress.

The Speech-Based Reflective Support Module aims to give students greater self-awareness and emotional control by using voice-based journaling. This module allows users to record their reflections in voice form, whereupon the system parses speech patterns to derive both textual information and inherent emotional cues. Detected emotions are subsequently represented through personalized mind maps, which enable students to monitor emotional changes and identify possible stress factors over time. To enhance peer support, the module also integrates with the Community Center, where students experiencing similar emotions can connect, share insights, and engage in moderate discussions. The implementation plan includes:

- i. **Audio Recording:** Capturing user voice inputs in a secure and private manner.
- ii. **Text & Emotion Detection:** Processing voice signals to detect emotional patterns.
- iii. **Mind Map Visualization:** Presenting emotional insights through interactive visualizations for easy reflection.
- iv. **Comparative Analysis:** Matching patterns with similar emotional profiles to provide personalized, reflective feedback.

This component contributes to the project by combining AI-driven emotional analysis with peer-based community support, ultimately promoting stress awareness, resilience, and mental well-being among students.

2. BACKGROUND & LITERATURE SURVEY

Emotion recognition is an important topic in human-computer interaction. People can understand others' emotions through facial expressions, speech content, and voice tone. Voice is especially important when face-to-face communication is not possible, such as in virtual meetings. In educational settings, being able to detect changes in students' emotions during their daily studies and life is also crucial for teachers to provide better support and guidance [3].

Speech emotion recognition (SER) can be approached in two main ways. Scenario I processes raw speech signals by extracting features like MFCCs, pitch, and energy, which are then classified using machine learning or deep learning models. Scenario II converts speech into visual forms, such as spectrograms, and analyzes them with image-based models like CNNs. Scenario I captures temporal dynamics but requires careful feature engineering, while Scenario II leverages image-based deep learning but often ignores speech's temporal information. Both approaches have limitations, motivating hybrid methods that combine spatial and temporal features for better emotion recognition.

Previous research has explored the use of Convolutional Neural Networks (CNN) for speech emotion classification, demonstrating its efficiency in detecting multiple emotions including annoyance, astonishment, unhappiness, calmness, nervousness, and joy. The study highlights a methodological basis for incorporating deep learning techniques to analyze emotional cues from speech, which could be adapted to enhance real-time emotion detection in interactive systems like virtual reality training environments [4].

Previous research on speech emotion recognition has explored a deep learning model combining 1D-CNN and GRU to detect emotions from speech for enhancing virtual assistants and mental health monitoring. Evaluated on the TESS dataset, it accurately recognizes seven emotions, adapts assistant responses, and logs emotional data for medical review. The model is efficient, requires minimal preprocessing, and outperforms existing methods, providing a practical solution for emotion-aware applications [5].

A study on Sentimental Voice Recognition (SVR) highlights the use of AI and machine learning to detect human emotions such as happiness, sadness, anger, and fear through speech analysis. Studies have explored techniques including feature extraction from pitch, tone, and rhythm, deep learning on audio-visual datasets, and human-robot interaction to classify emotional states with high accuracy. Building on these works, the proposed SVR model collects diverse voice datasets, preprocesses and extracts relevant acoustic features, and applies machine learning algorithms for emotion classification. This approach has shown effectiveness in real-time applications, including intelligent

call centers, vehicle safety, and interactive robotic systems, with potential for future enhancements to detect additional emotions and integrate alert mechanisms for critical emotional states [6].

Another relevant study explored real-time recognition of affective states from nonverbal features of speech to support public speaking skill analysis. The researchers developed a system that extracts various acoustic features from a speaker's voice, such as pitch, energy, and speech rate, and applies machine learning models to detect stress, anxiety, and confidence levels. Unlike conventional post-event analysis, their approach emphasizes real-time feedback, allowing speakers to adjust their performance dynamically. The study demonstrated that integrating machine learning with real-time voice analysis can effectively detect emotional states during public speaking, offering a foundation for training systems that enhance performance and reduce communication-related anxiety [7].

Existing literature has highlighted the use of mobile applications in supporting student stress management. Researchers have investigated solutions that rely on standardized questionnaires, such as Anxiety, Depression, and Stress Scale (ADSS), to evaluate mental health levels and recommend relaxation strategies [8].

Previous research has also demonstrated the effectiveness of integrating journaling within mobile-based stress management interventions. For example, [9] incorporated features such as guided relaxation, breathing exercises, and a journaling function where users recorded reflections before and after stress-reduction activities. Findings from a study involving secondary school students indicated that the journaling component, alongside therapeutic exercises, supported adolescents in tracking their emotional states and reflecting on their stress levels, contributing to overall reductions in perceived stress [10]. This highlights the vital role of journaling in fostering self-awareness and emotional regulation within mobile stress management systems, a direction that aligns closely with the aims of the current study, which emphasize voice-based journaling for deeper emotional insight and reflection.

Previous research has highlighted the importance of personalization in mobile health applications. [11] that allows medical specialists to design personalized stress management applications without requiring programming expertise. Their system integrates sensors, questionnaires, data processing, treatment recommendations, and feedback, enabling tailored stress interventions delivered through mobile devices. The study demonstrated the feasibility of implementing a stress application where heartbeat monitoring and questionnaires informed stress diagnosis and coping suggestions. This emphasizes the effectiveness of personalized, mobile-based stress solutions, supporting the idea that adaptive and user-specific systems, such as voice-based journaling and emotional trend mapping, can provide meaningful improvements over generic stress management approaches.

Another study emphasized the potential of mobile health applications in managing stress and supporting emotional regulation. [12] conducted a systematic review of 11 studies focused on work-related stress and found that mHealth interventions, particularly mindfulness-based meditation programs, were effective in reducing perceived stress, burnout, and anxiety, while enhancing well-being and coping skills. Importantly, the review also highlighted features such as daily diaries and emotion journaling as valuable tools to promote self-awareness and emotional self-regulation, demonstrating that journaling plays a vital role in sustaining engagement and tracking progress in stress management. These findings support the integration of voice-based journaling and emotional trend visualization in academic stress interventions, offering a more dynamic and personalized approach.

Recent studies have examined the opportunities and challenges of mobile mental health applications. [13] analyzed over 3,000 user reviews from leading mental health apps and found that features such as mood tracking and journaling were among the most valued tools for supporting self-monitoring and emotional regulation. However, their findings also revealed significant challenges, including app reliability issues, high attrition rates, and user frustration with repetitive or intrusive features. Importantly, while journaling and tracking were positively associated with self-awareness, many users still reported unmet needs, particularly those with conditions like PTSD, ADHD, and bipolar disorder. These insights highlight both the critical role of journaling in stress management and the need for more adaptive, personalized, and reliable tools, which justify the integration of voice journaling and emotion-based self-reflection in academic stress management.

3. RESEARCH GAP

Previous studies emphasize the need to validate stress assessment tools on larger student populations, improve app usability through user-centered design, and integrate AI for personalized stress management among students. However, limited research exists on developing computational models to predict stress onset and evaluate their effectiveness through longitudinal studies [14, 9]. Through voice journaling, a personalized mind map can be developed for each student, capturing their unique stressors, coping mechanisms, and emotional patterns. This approach not only provides deeper insights into individual experiences but also supports the development of tailored interventions. By transforming unstructured voice inputs into structured, visual representations, students can better understand their thoughts and behaviors.

Although existing applications provide stress assessments and relaxation techniques, few integrate continuous tracking of assessment results with community-based peer support [15]. There is limited research on systems that allow students to view their previous assessment levels, receive personalized recommendations on re-login, and simultaneously engage with a community center for sharing experiences. This gap highlights the need for solutions that combine personalized progress monitoring with collaborative peer support in managing academic stress.

Another study highlighted the importance of developing a computerized “relaxation program” that could be integrated into an anti-stress application [16]. Such an approach would be especially valuable for individuals who are unlikely to seek in-person medical support. Through voice journaling, users can better understand their stress levels, while also engaging with a supportive community to help reduce stress.

4. RESEARCH PROBLEM

University students face increasing levels of academic stress, yet existing digital interventions often lack personalization, continuous monitoring, and peer support integration. While many applications provide one-time assessments or generic relaxation techniques, few systems combine adaptive stress tracking, voice journaling for self-reflection, and community-based support. This gap limits the effectiveness of digital tools in addressing the dynamic and multifaceted nature of student stress and reduces opportunities for personalized stress management strategies.

5. OBJECTIVES

5.1 Main Objectives

To provide a comprehensive digital solution for academic stress management by integrating voice journaling, adaptive stress heatmaps, AI-driven emotional support, personalized daily assessments, and a community discussion forum, thereby enhancing self-awareness, resilience, and social support to improve students' mental well-being and academic performance.

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5.2 Sub Objectives

- i. **To design the voice journalling system architecture:** To design system architecture, interface models, and UML diagrams ensuring usability, scalability, and performance.
- ii. **To develop a voice journaling module:** To enable users to record their thoughts, reflections, and experiences verbally, with the recorded voice data transcribed into text for further analysis of emotional patterns and user engagement.
- iii. **To implement voice-based emotion and key text recognition:** To identify emotional states such as stress, sadness, or calmness using acoustic features, and to extract emotionally significant words and phrases from transcribed journal entries for a comprehensive understanding of users' affective conditions in real time.
- iv. **To generate personalized emotional theme maps:** To link detected emotional tones from voice recordings to user-specific topics and life themes, helping users understand which aspects of their academic and personal life contribute to their emotional states.
- v. **To integrate a comparative insight module:** To compare an individual's emotional expressions with anonymized profiles of similar users, enhancing self-awareness and providing benchmarks for emotional regulation and stress management.
- vi. **To facilitate self-reflection and monitoring:** To provide users with visual feedback on emotional patterns and stress triggers, promoting conscious monitoring of mental well-being, early intervention, and the adoption of personalized coping strategies.

6. METHODOLOGY

This study focuses on developing a Voice journaling component that helps students monitor and manage stress using voice-based interactions. The system allows students to record voice journals, detect stress-related keywords, analyzes emotions, and provides personalized relaxation suggestions. It also includes a community feature where students can share experiences and support each other.

6.1 System Diagram

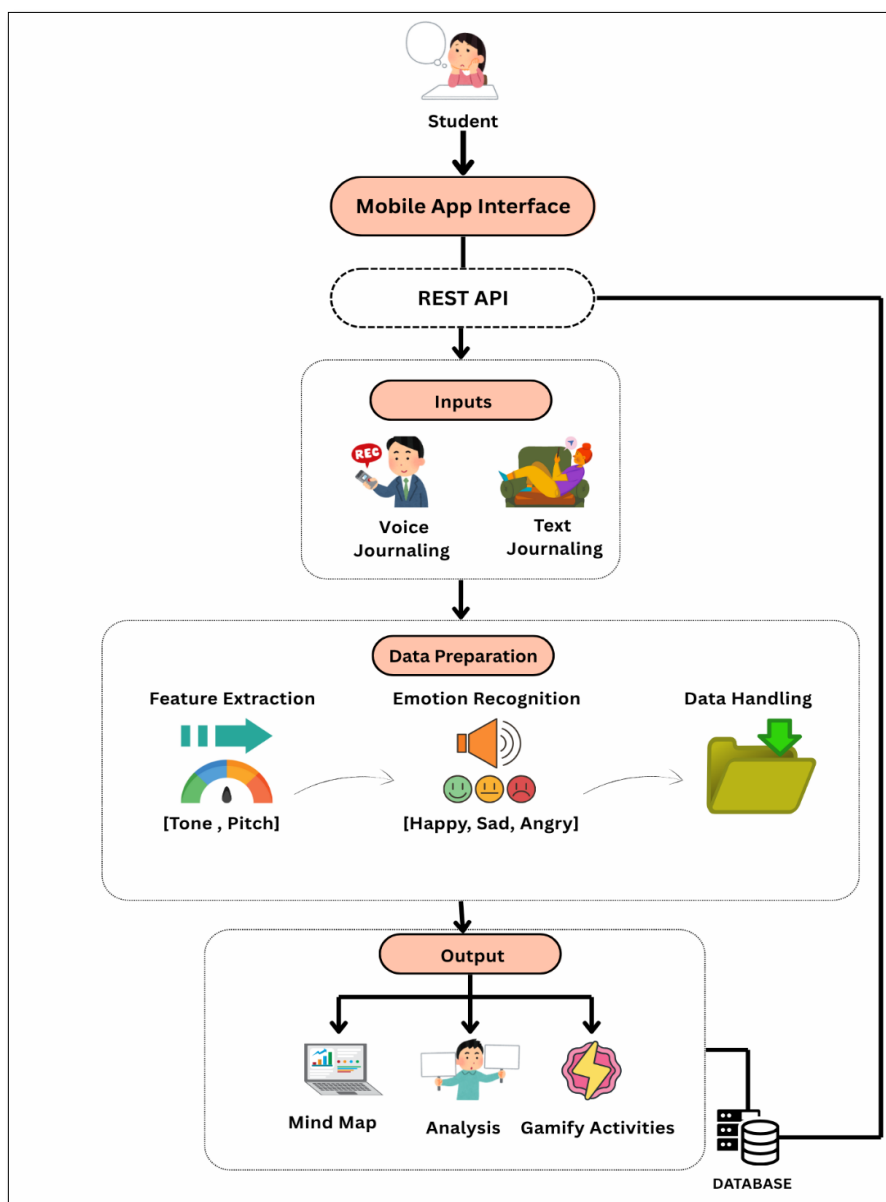


Figure 1-System Conceptual Diagram

6.2 Data Collection and Preprocessing

Public Speech Emotion Datasets

- i. RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song)
- ii. Emo-DB (Berlin Database of Emotional Speech)
- iii. Speech Emotion Recognition

6.3 Preprocessing steps

- i. Noise Removal: Eliminate background sounds and unwanted disturbances to focus solely on the student's voice.
- ii. Normalization: Standardize volume levels across all recordings to reduce differences in loudness that may affect model performance.
- iii. Data Augmentation: Apply techniques such as pitch shifting, speed variation, and time stretching to increase dataset diversity and improve model robustness.
- iv. Quality Assurance: Ensure all preprocessing steps enhance the accuracy and reliability of keyword spotting and emotion recognition modules.

6.4 Model Training and Implementation

- i. Keyword Spotting (KWS) Model
- ii. The preprocessed audio is used to train a model that identifies stress-related keywords. Both real and synthetic (TTS-generated) recordings are used to ensure the model can recognize speech patterns from different students.
- iii. Emotion Recognition Model
- iv. Extracted features (MFCCs, pitch, energy) from preprocessed audio are fed into a Convolutional Neural Network (CNN) or other suitable machine learning models. The model classifies emotions such as stress, calmness, sadness, nervousness, and joy. Synthetic voices are adjusted to match real student speaking styles for improved accuracy.
- v. Integration with Application
- vi. Trained models are integrated into the mobile application. The system combines KWS and emotion recognition outputs to assess a student's current stress level.

Stress scores are stored in the database, and the application provides personalized feedback and relaxation interventions.

After model training, the application workflow begins with the student recording a voice journal through the app. The recorded audio is then processed through the preprocessing pipeline, which includes noise removal, normalization, and feature extraction. The keyword spotting (KWS) module analyzes the processed audio to identify stress-related keywords, while the emotion recognition module determines the student's emotional state. Based on the combined outputs, the system provides personalized mind map tailored to the student's stress level. Students can engage with the community module to share experiences and receive support from peers, enhancing the overall effectiveness of the stress management process.

6.5 Tools and Technologies

The Academic Stress Releaser Application is developed using a combination of modern frontend, backend, and machine learning technologies to ensure functionality, scalability, and efficient model implementation.

i. Frontend

React Native is used to develop mobile applications, providing a cross-platform interface compatible with both Android and iOS devices. The UI allows students to record voice journals, view stress levels, access relaxation exercises, and interact with the community module.

ii. Backend and Database

Firebase is employed for backend services, including authentication, real-time database management, and cloud storage for voice recordings and user data. Firebase ensures secure data handling and enables real-time updates for community interactions and stress tracking.

iii. Model Implementation

Machine Learning (ML) and Deep Learning (DL) techniques are utilized to implement the Keyword Spotting (KWS) and Emotion Recognition modules. Python is used for model development. Feature extraction from audio signals, such as MFCCs, pitch, and energy, is performed using libraries like Librosa. Text-to-Speech (TTS) systems are integrated to generate synthetic voice data for model training and augmentation, improving model accuracy and robustness.

iv. Version Control

To ensure proper management of the application code and facilitate collaboration among team members, Git is used as the version control system, with GitHub serving as the remote repository. Git enables tracking of changes in the codebase, management of multiple development branches, and efficient merging of updates from different contributors. GitHub provides a centralized platform for storing the project repository, documenting progress, and managing issues. By employing GitHub, the development process becomes more organized, collaborative, and secure, allowing for easy rollback to previous versions in case of errors and ensuring consistent integration of new features.

7. PROJECT REQUIREMENTS

7.1 Functional Requirements

- i. Voice Journal Recording- The app must allow students to record daily voice journals.
- ii. Keyword Spotting (KWS)- Detect stress-related words or phrases in the user's speech.
- iii. Emotion Recognition- Analyze the user's speech to classify emotions such as stress, calmness, sadness, nervousness, and joy.
- iv. Stress Level Calculation- Calculate and store stress score based on detected keywords and emotions.
- v. Relaxation Intervention- Provide personalized relaxation exercises or coping strategies based on stress levels.
- vi. Community Interaction- Allow users to post messages, share experiences, and provide peer support.
- vii. Data Storage & Retrieval- Store voice recordings, stress scores, and user information securely in a database. Allow retrieval of historical data for tracking stress trends over time.

7.2 Non-Functional Requirements

- i. Performance- The app should process audio recordings and return stress results quickly. It must respond within a few seconds to ensure smooth user experience.
- ii. Scalability- The system should handle multiple users simultaneously. Performance should remain stable even with increased user load.
- iii. Reliability- All voice recordings, stress scores, and community data must be saved securely. Users should be able to retrieve their data without loss.
- iv. Usability- The interface must be simple, intuitive, and easy to navigate. Students should be able to use the app without prior training.

- v. Security & Privacy- User data and recordings must be encrypted and protected. Access should be restricted to authorized users only.
- vi. Maintainability- The codebase should follow best practices for easy updates and bug fixes. New features should be added without affecting existing functionality.
- vii. Portability- The application must run smoothly on both Android and iOS devices. It should be adaptable to different device specifications and screen sizes.

7.3 User Requirements

- i. Users must be able to record and save their daily voice journals easily with a single tap.
- ii. Users should receive instant feedback on their stress level after submitting a recording.
- iii. Users should be able to view their past recordings and stress history in a timeline format.
- iv. The app should provide personalized relaxation tips or exercises, such as breathing exercises or meditation guides, based on the detected stress level.
- v. Users must be able to interact with the community by sharing supportive messages and personal experiences.
- vi. The system should suggest peers with similar stress patterns so that users can optionally connect with them.
- vii. Users must be able to accept or reject community connection requests to maintain their privacy.
- viii. Users should be able to log in securely and manage their personal profile.
- ix. Users should receive non-intrusive notifications or reminders that encourage them to record voice journals regularly.

7.4 Test Cases

To ensure the system functions as expected, test cases were designed based on the functional requirements. These test cases validate whether each feature meets the intended outcome and provide a foundation for quality assurance.

Module Name	Voice Journaling		
Test Case ID	VJ_TC_01		
Test Case Description	Verify that the system records and transcribes a user's voice journal entry correctly.		
Prerequisites	Users are logged into the system with microphone permission enabled.		
Environmental Information	Mobile apps, Firebase backend, stable internet connection.		
Test Scenario	User records a 1-minute voice journal.		
Test Steps	Test Input	Expected Results	Comments
Navigate to Voice Journaling Module.	—	Voice journaling screen is displayed.	UI validation
Click "Start Recording."	Tap on record button.	Recording starts with timer visible.	—
Speak journal entry for 1 minute.	"I am stressed because of upcoming exams."	System saves audio file & initiates transcription.	—
Stop recording.	Tap stop button.	System saves audio file & initiates transcription.	—
View transcription.	—	Transcribed text displayed: <i>"I am stressed because of upcoming exams."</i>	Accuracy within 95%

Figure 2 - Test Case 1_Voice Journaling

Module Name	Voice Journaling - Emotion Recognition & Mapping		
Test Case ID	VJ_TC_02		
Test Case Description	Verify that the system analyzes tone, detects emotion, and generates emotional theme map.		
Prerequisites	User has at least one transcribed journal entry stored in the system.		
Environmental Information	Mobile app with NLP & emotion recognition API enabled.		
Test Scenario	System processes a saved journal entry to detect emotions and link themes.		
Test Steps	Test Input	Expected Results	Comments
Select an existing journal entry.	“I am stressed because of upcoming exams.”	Entry loads in the system.	–
Run emotion analysis.	–	System detects “ <i>stress / anxiety</i> ” emotion.	Accuracy check
Apply NLP keyword extraction.	–	System identifies keywords: “ <i>exams, stress.</i> ”	Topic mapping
Generate emotional theme map.	–	Visual mind map generated linking “ <i>exam</i> ” to “ <i>stress/anxiety.</i> ”	Validate mapping UI

Figure 3- Test Case 2_ Voice Journaling - Emotion Recognition & Mapping

Module Name	Voice Journaling - Community Suggestions		
Test Case ID	VJ_TC_03		
Test Case Description	Verify that after a user records a voice journal, the system suggests community		
Prerequisites	– User is logged in. – At least 10 community members exist with stored journals. – Journals are indexed for similarity (keywords/emotion tags).		
Environmental Information	Mobile application, NLP/emotion analysis module, Firebase/ML model for matching.		
Test Scenario	A user records a new voice journal entry. The system analyzes the content, suggests relevant community members, and enables direct messaging.		
Test Steps	Test Input	Expected Results	Comments
Navigate to <i>Voice Journal</i> .	–	Voice journaling interface opens.	–
Record and submit a journal.	Voice input: <i>“I feel stressed before exams and can’t sleep properly.”</i>	Journal saved in database.	–
System processes entry.	NLP/Emotion analyzer parses keywords (e.g., “exam stress,” “sleep”).	Tags created: <i>stress, exam, sleep</i> .	Processing validation
Matching algorithm runs.	Compare tags with other community journals.	System identifies members with similar tags.	–
Suggestions displayed.	–	List of community members (e.g., <i>3 peers with exam-related stress</i>) shown to user.	Accuracy check
User chooses to connect.	Tap on suggested member.	“Message” option appears.	–
User sends a message.	Text: <i>“Hi, I’m also struggling with exam stress. Can we talk?”</i>	Message delivered to selected community member.	Final validation

Figure 4- Test Case 3_Voice Journaling - Community Suggestions

7.5 Sampl Sketch of the Proposed System



Figure 5-Sketch_Voice Journaling

8. WORK BREAK DOWN STRUCTURE

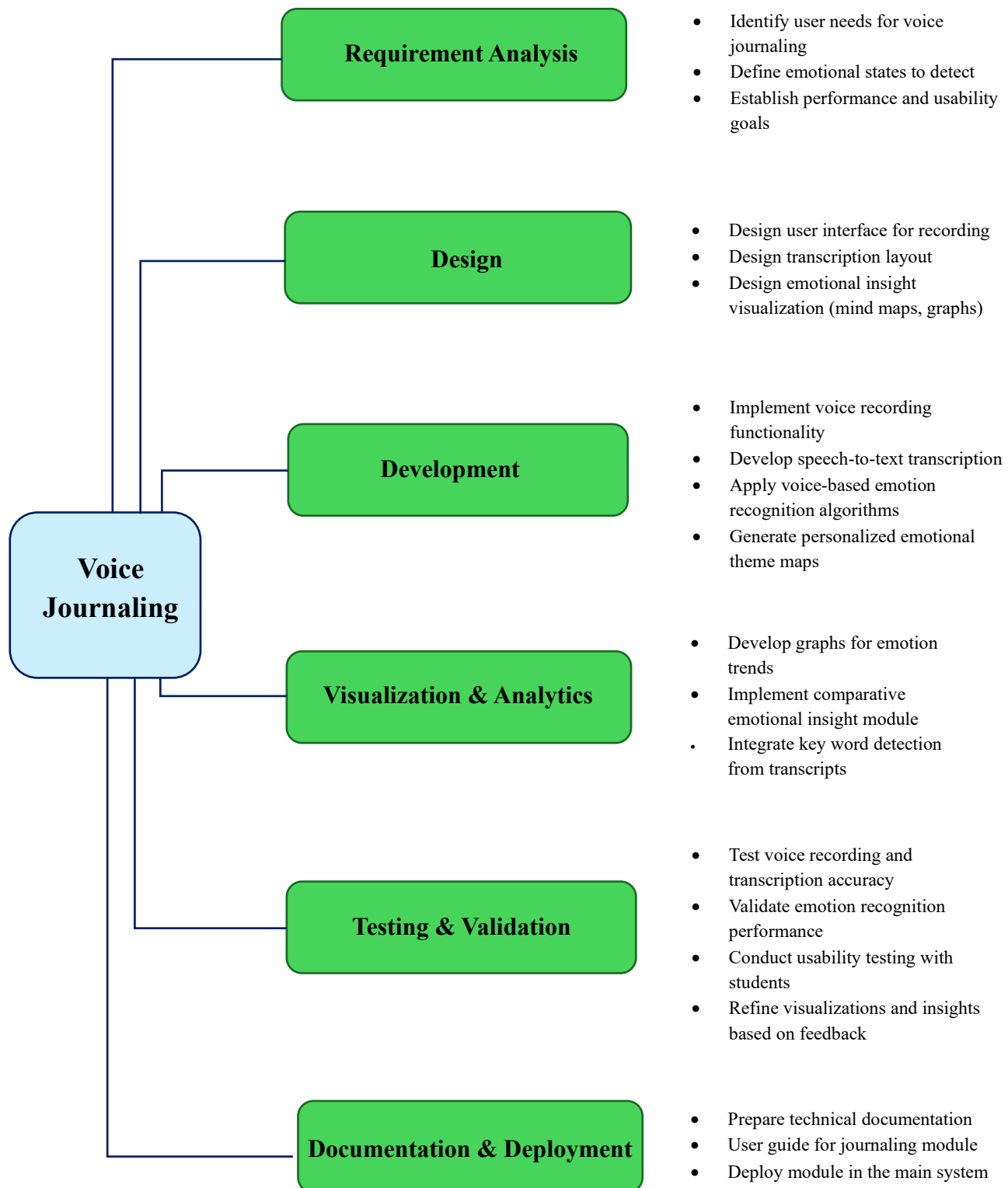


Figure 6-Work Breakdown Structure

9. GANTT CHART

Project Phase	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7
Project Planning & Requirements							
System Design							
Data Collection & Preprocessing							
Model Development (KWS & Emotion)							
Application Development (Frontend & Backend)							
System Integration & Testing							

Figure 7- Gantt Chart

10. BUDGET AND BUDGET JUSTIFICATION

Category	Item/Service	Quantity	Unit Cost (LKR)	Total Cost (LKR)
Hardware	Laptop/Mobile Phone/Other equipment	2	250,000	500,000
Software/Tools	Firebase Cloud Services (Hosting, DB)	1 Year	30,000	30,000
	Voice Emotion Recognition API	1 Year	50,000	50,000
Human Resources	External Supervision (Hiring Professionals in relevant field)	1 Year	25,000	150,000
Services	Domain & Hosting	1 Year	15,000	15,000
Miscellaneous	Data Collection & Testing Tools		20,000	20,000
Other Cost	Transportation Cost	1 Year	5000	60,000
	Stationary Cost	1 Year	4000	48,000
Total				873,000

i. **Hardware LKR 500,000**

Two laptops or mobile devices are required to support development, testing, and deployment of the voice journaling system. This ensures that multiple team members can work concurrently and that testing covers different device configurations for accurate functionality.

ii. **Software/Tools LKR 80,000**

- **Firestore Cloud Services LKR 30,000-** Used for secure data storage, real-time database management, and hosting the application. It ensures scalability and seamless data access.

- **Voice Emotion Recognition API LKR 50,000-** Essential for extracting emotional features from users' voice recordings, enabling real-time emotion detection and personalized feedback.
- iii. **Human Resources LKR 150,000**
External supervision involves hiring professionals or consultants with expertise in emotion recognition, machine learning, or psychology. Their guidance ensures the system's algorithms are scientifically valid and reliable over the course of the project.
- iv. **Services LKR 15,000**
Domain registration and hosting are necessary for deploying the application in a secure and accessible online environment for student users.
- v. **Miscellaneous LKR 20,000**
Data collection and testing tools include software, microphones, and other instruments needed to gather high-quality voice samples and validate emotion recognition performance.
- vi. **Other Costs LKR 108,000**
 - **Transportation LKR 60,000-** For field visits, user testing sessions, or data collection activities.
 - **Stationery LKR 48,000-** Includes printing, documentation, and other consumables required throughout the project lifecycle.

Total Budget LKR 873,000

This budget ensures the successful development, testing, and deployment of the voice journaling module, providing a reliable and scientifically validated system for academic stress management.

11. COMMERCIALIZATION STRATEGIES

11.1 Commercialization and Revenue Opportunities

- i. **Freemium Model:** Offer a basic version of the app for free, with core features like voice journaling and trend visualization. Premium upgrades unlock advanced functionalities, such as extended storage of past journals (up to six months), institutional guidance, and larger community access (more than 10 members).
- ii. **Institutional Partnerships:** Collaborate with universities, colleges, and counseling centers to provide the app as part of student wellness programs. Institutions can subscribe to premium features to monitor aggregate trends and provide tailored support.
- iii. **Subscription Plans:** Introduce monthly or yearly subscription plans for premium users who want access to advanced analytics and extended community interactions.

11.2 User Opportunities Compared to Traditional Counseling

- i. **Flexible Access:** Users can engage with the platform anytime and anywhere, removing scheduling and location constraints of in-person counseling.
- ii. **Anonymity & Comfort:** Students can express their emotions privately without fear of stigma, which may encourage more honest self-reflection.
- iii. **Continuous Self-Monitoring:** Voice journaling with trend visualization allows students to track emotional patterns over time, identify stress triggers early, and implement coping strategies proactively.
- iv. **Peer & Institutional Support:** Through community forums and premium institutional guidance, users gain social and professional support that complements personal reflection, providing a holistic support system beyond what traditional counseling alone offers.

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